

E-Commerce Sales Analytics Dashboard

SQL • Python • Power BI • ETL • Data Visualization

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Project Overview

An end-to-end data analytics project analysing **100,000+ real e-commerce orders** from the Olist Brazilian E-Commerce dataset. This project covers the complete Data Analyst workflow — raw data ingestion, SQL analysis, Python ETL pipeline, and interactive Power BI dashboard — uncovering actionable business insights across revenue, customer behaviour, product performance, and logistics.

Business Questions Answered

#	Business Question	Insight Found
1	How did revenue trend over time?	Revenue grew 12x in 7 months (Oct 2016 → May 2017)
2	Which product categories drive the most revenue?	Beauty & Health (beleza_saude) leads at R\$1.26M
3	Which states have the highest average order value?	PB (Paraíba) leads at R\$248 avg order value
4	How has the cancellation rate changed over time?	Dropped from 7.4% (2016) to under 1.5% (2017+)
5	What is the customer retention rate?	Only 3.12% repeat buyers — critical retention problem
6	Does delivery time affect customer satisfaction?	1-star orders take 2x longer (20.8 days vs 10.2 days)

#	Business Question	Insight Found
7	Which payment methods drive the most revenue?	Credit card dominates at R\$12.54M (77% of revenue)
8	What are the top revenue months?	November 2017 (Black Friday) = highest at R\$1.19M

Key Insights

Delivery time is the #1 driver of customer satisfaction. Orders delivered within 10 days average 5-star reviews. Orders taking 20+ days average 1-star reviews — a 50% drop in satisfaction.

96.88% of customers never return. The business has a severe retention problem — nearly all revenue depends on acquiring new customers rather than retaining existing ones.

Beauty & Health is the star category. beleza_saude generates R\$1.26M — 27% more than the second highest category.

Tools & Technologies

Tool	Purpose
SQL (SQLite)	Data extraction, transformation, business queries
Python (Pandas)	ETL pipeline, CSV export automation
Power BI	Interactive dashboard, DAX measures, data modelling
GitHub	Version control and portfolio hosting

Project Structure

```
ecommerce-sales-analytics/  
|  
├── README.md  
|  
└── sql/
```

```
|   └─ queries.sql                # All 8 business SQL queries
|
└─ python/
|   └─ export.py                  # ETL pipeline - loads CSVs, runs queries,
exports results
|
└─ dashboard/
|   └─ ecommerce_sales_dashboard.pbix  # Power BI dashboard file
|
└─ data/
|   └─ monthly_revenue.csv
|   └─ top_categories.csv
|   └─ revenue_by_state.csv
|   └─ cancellation_rate.csv
|   └─ customer_retention.csv
|   └─ delivery_vs_reviews.csv
|   └─ payment_types.csv
|   └─ top_revenue_months.csv
|
└─ screenshots/
    └─ dashboard.png              # Full dashboard screenshot
```

Dashboard Visuals

Visual	Type	Insight
Monthly Revenue Trend	Line Chart	12x revenue growth from 2016 to 2018
Top 10 Product Categories	Bar Chart	Beauty & Health dominates
Order Cancellation Rate	Area Chart	Dramatic improvement from 7.4% to under 1%
Repeat vs One-Time Buyers	Donut Chart	96.88% one-time buyers — retention crisis
Delivery Days vs Review Score	Column Chart	Slower delivery = lower ratings
Revenue by Payment Type	Treemap	Credit card = 77% of all revenue

Slicers: Filter by Month • Filter by Category • Filter by Payment Type

Dataset

- **Source:** Olist Brazilian E-Commerce Dataset — Kaggle
 - **Size:** 100,000+ orders across 9 tables
 - **Period:** September 2016 — October 2018
 - **Tables used:** olist_orders, olist_order_items, olist_products, olist_customers, olist_order_payments, olist_order_reviews
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How to Run

1. Download the Olist dataset from Kaggle
 2. Place all CSV files in the project folder
 3. Run `python/export.py` to generate all 8 analysis CSV files
 4. Open `dashboard/ecommerce_sales_dashboard.pbix` in Power BI Desktop
 5. Refresh data source to point to your local CSV files
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Author

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